Note that the set of all natural numbers, the set of all non-negative integers, set of all odd integers, set of all even integers, set of all prime numbers are not intervals. Further observe that, between any two real numbers there are infinitely many real numbers and hence the above sets are not intervals.

Consider the following sets:

(i) The set of all real numbers greater than 0.
(ii) The set of all real numbers greater than 5 and less than 7.
(iii) The set of all real numbers x such that $1 \leq x \leq 3$.
(iv) The set of all real numbers x such that $1 < x \leq 2$.

The above four sets are intervals. In particular (i) is an infinite interval and (ii), (iii) and (iv) are finite intervals. The term "finite interval" does not mean that the interval contains only finitely many real numbers, however both ends are finite numbers. Both finite and infinite intervals are infinite sets. The intervals correspond to line segments are finite intervals whereas the intervals that correspond to rays and the entire real line are infinite intervals.

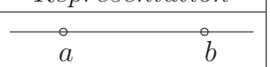
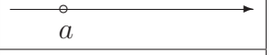
A finite interval is said to be ***closed*** if it contains both of its end points and ***open*** if it contains neither of its end points. Symbolically the above four intervals can be written as $(0, \infty), (5, 7), [1, 3], (1, 2]$. Note that for symbolic form we used parentheses and square brackets to denote intervals. ( ) parentheses indicate open interval and [ ] square brackets indicate closed interval. The first two sets are open intervals, third one is a closed interval. Note that fourth set is neither open nor closed, that is, one end open and other end closed.

In particular $[1, 3]$ contains both 1 and 3 and in between real numbers. The interval $(1, 3)$ does not contain 1 and 3 but contains all in between the numbers. The interval $(1, 2]$ does not contain 1 but contains 2 and all in between numbers.

Note that ∞ is not a number. The symbols $-\infty$ and ∞ are used to indicate the ends of real line. Further, the intervals (a, b) and $[a, b]$ are subsets of $\mathbb{R}$.

Type of Intervals

There are many types of intervals. Let $a, b \in \mathbb{R}$ such that $a < b$. The following table describes various types of intervals. It is not possible to draw a line if a point is removed. So we use an unfilled circle "$\circ$" to indicate that the point is removed and use a filled circle "$\bullet$" to indicate that the point is included.

Interval	*Notation*	*Set*	*Diagramatic Representation*
finite	(a, b)	$\{x : a < x < b\}$	
	$[a, b]$	$\{x : a \leq x \leq b\}$	
	$(a, b]$	$\{x : a < x \leq b\}$	
	$[a, b)$	$\{x : a \leq x < b\}$	
infinite	(a, ∞)	$\{x : a < x < \infty\}$	
	$[a, \infty)$	$\{x : a \leq x < \infty\}$	
	$(-\infty, b)$	$\{x : -\infty < x < b\}$	
	$(-\infty, b]$	$\{x : -\infty < x \leq b\}$	
	$(-\infty, \infty)$ or $\mathbb{R}$	$\{x : -\infty < x < \infty\}$ or the set of real numbers	